

REVIEW

Efficacy and Safety of All Monoclonal Antibodies in Moderate-to-Severe Atopic Dermatitis: A Systematic Review and Network Meta-Analysis

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Received: 21 April 2025 | **Revised:** 25 September 2025 | **Accepted:** 5 November 2025

Keywords: atopic dermatitis | efficacy | monoclonal antibodies | network meta-analysis | randomized controlled trials | safety

ABSTRACT

Objective: The aim of this study was to compare the efficacy and safety of monoclonal antibodies (mAb) for moderate-to-severe atopic dermatitis (AD).

Methods: The randomized controlled trials (RCTs) of monoclonal antibodies in the treatment of moderate-to-severe AD were searched in the database of PubMed, Embase, Web of Science and Cochrane Library, up to November 2024. The control group included placebo. The efficacy indicators were the percentage of patients achieving 50%, 75%, 90% improvement in Eczema Area and Severity Index score (EASI-50, EASI-75, EASI-90) and the percentage of patients with an Investigator Global Assessment (IGA) Score of 0 or 1 from baseline until the time of efficacy observation, and the percent change in Pruritus Numerical Rating Scale (NRS), EASI score, SCORing Atopic Dermatitis (SCORAD), and change in Percent Body Surface Area (BSA), Dermatology Life Quality Index (DLQI). The statistical analysis was performed by Stata14 and RevMan5.4. Data processing, network evidence plots, surface under the cumulative ranking curve (SUCRA) ranking, league plots and funnel plots were generated. Risk ratio (RR) and 95% confidence interval (95% CI) were used as effect sizes to analyze binary categorical variables.

Results: This study included 32 RCTs with 7588 patients. Spesolimab, Rademikibart, Dupilumab, Amltelimab were more effective than placebo in EASI-50 (RRs ranging between 1.31 and 22.65), EASI-75 (RRs ranging between 1.51 and 36.58), EASI-90 (RRs ranging between 3.72 and 5.49), and the percentage of patients in IGA score of 0 or 1 (RRs ranging between 1.78 and 13.36). Dupilumab showed relatively good efficacy according to SUCRA ranking on percent change in NRS, EASI, SCORAD. Dupilumab exhibited a lower incidence of serious adverse events than placebo, and the difference was statistically significant (RR = 0.43, 95% CI 0.30–0.63). Other safety analysis results showed no statistical difference.

Conclusions: Through the analysis of the primary efficacy indicators, this network meta-analysis (NMA) study indicated that all monoclonal antibodies performed better than placebo. Based on the results of this study, Spesolimab, Rademikibart, Dupilumab, Amltelimab were recommended treatment options with relatively good efficacy and safety.

1 | Introduction

Atopic dermatitis (AD) is a common chronic inflammatory skin disease, with a prevalence of 25%–30% in children and 10% in adults

[1–3]. It is characterized by intense pruritus, recurrent eczema, skin barrier dysfunction and type II inflammation [4]. These manifestations often lead to skin pain, sleep disturbances, increased risk of infections, and substantially impaired quality of life [5–7].

Summary

- This study is the first to comprehensively evaluate all clinically approved monoclonal antibodies for the treatment of moderate-to-severe atopic dermatitis (AD), filling a gap in the existing literature.
- A network meta-analysis (NMA) approach was used, providing a broader comparative perspective on treatment efficacy than traditional meta-analyses.
- The study included data up to November 2024, ensuring the results reflect the latest advancements in the field.
- High-quality randomized controlled trials (RCT) were included, and statistical methods such as SUCRA ranking were employed to ensure the reliability of the results.
- The findings provide evidence-based treatment options for clinicians, helping to improve treatment decisions and patient care.

Plain Language Summary

This study looked at how effective and safe different monoclonal antibodies are for treating moderate-to-severe atopic dermatitis (AD), a chronic skin condition that causes itching, redness, and inflammation. We reviewed 32 clinical trials involving 7588 patients to compare monoclonal antibodies against placebo (a non-active treatment). The results showed that monoclonal antibodies like Spesolimab, Rademikibart, Dupilumab, and Amlitelimab were significantly better than placebo in improving the severity of AD, as measured by different skin scores and patient-reported outcomes. Dupilumab, in particular, had strong effects on reducing itching and improving skin conditions. It was also associated with fewer serious side effects compared to placebo. However, other monoclonal antibodies did not show significant differences in safety. Overall, this study suggests that monoclonal antibodies are a promising treatment option for people with moderate-to-severe AD, with Dupilumab standing out for its effectiveness and safety.

Treatment of AD is stratified by disease severity. Mild to moderate cases are generally managed with topical corticosteroids, calcineurin inhibitors, or phototherapy [8–10]. However, patients with moderate-to-severe disease frequently require systemic therapy. Conventional immunomodulators such as cyclosporine, azathioprine, methotrexate, and mycophenolate have been used [11, 12], albeit with limitations due to their safety profiles and suitability only for short-term use. For instance, cyclosporine—though approved for severe AD in some regions—is associated with nephrotoxicity and hepatotoxicity [13]. More recently, Janus kinase (JAK) inhibitors (e.g., abrocitinib, baricitinib, upadacitinib) have been introduced and show considerable efficacy [14–18]. However, long-term safety remains a consideration with these agents.

Over the past 8–10 years, monoclonal antibodies (mAb) have emerged as targeted therapeutic options for moderate-to-severe AD. Dupilumab, which inhibits IL-4 and IL-13 signaling, was

among the first mAb approved and has demonstrated robust efficacy and a favorable safety profile [19]. Other agents such as Tralokinumab and Lebrikizumab also target the IL-13 pathway [20–22], while Nemolizumab modulates IL-31 signaling, thereby addressing pruritus and inflammation [23–26]. More recent developments include mAb like Amlitelimab (anti-OX40L) [27–29], Ustekinumab (anti-IL-12/23) [30, 31], Fezakinumab (anti-IL-22) [32], Astegolimab (anti-IL-33) [33], and Spesolimab (anti-IL-36R) [34], all of which are under investigation or recently approved. These agents offer novel mechanisms of action and show promising outcomes in clinical trials.

Given the expanding repertoire of mAb for AD, there is a growing need to evaluate their relative efficacy and safety. Network meta-analysis (NMA) enables simultaneous comparison of multiple interventions, integrating direct and indirect evidence [35]. This systematic review and NMA aim to comprehensively assess all mAb used in moderate-to-severe AD, ranking their therapeutic and safety profiles to inform clinical decision-making.

2 | Materials and Methods

2.1 | Search Strategy

We systematically searched the PubMed, Embase, Web of Science, and Cochrane Library databases, spanning from their inception to November 2024, to ensure a comprehensive review of available literature. The following descriptors were used: “Amlitelimab”, “Astegolimab”, “CM310”, “Dupilumab”, “Eblasakimab”, “Fezakinumab”, “Lebrikizumab”, “Nemolizumab”, “Rademikibart”, “Spesolimab”, “Tralokinumab”, “Ustekinumab”, “Atopic Dermatitis”, “Randomized controlled trial”. Broaden the scope of the search to encompass references included in the database.

2.2 | Inclusion and Exclusion Criteria

The included studies were in English only and were RCTs. These patients received monotherapy with mAb, and trials in combination with corticosteroids were excluded. The control group was placebo. Trials that did not provide detailed information on EASI-50, EASI-75, EASI-90, or trials that did not include a placebo group were excluded.

Primary efficacy outcomes were EASI-50, EASI-75, EASI-90 and the percentage of patients with an IGA Score of 0 or 1 reflecting clear or mild AD. Secondary efficacy outcomes included the percent change in NRS, EASI score, SCORAD from baseline until the time of efficacy observation, and the change in BSA and DLQI. Safety outcomes mainly included adverse events (AE), serious adverse events (SAE), AE leading to discontinuation.

2.3 | Literature Selection and Data Collection

We organized the literature using NoteExpress software. Initially, two individuals conducted a preliminary screening

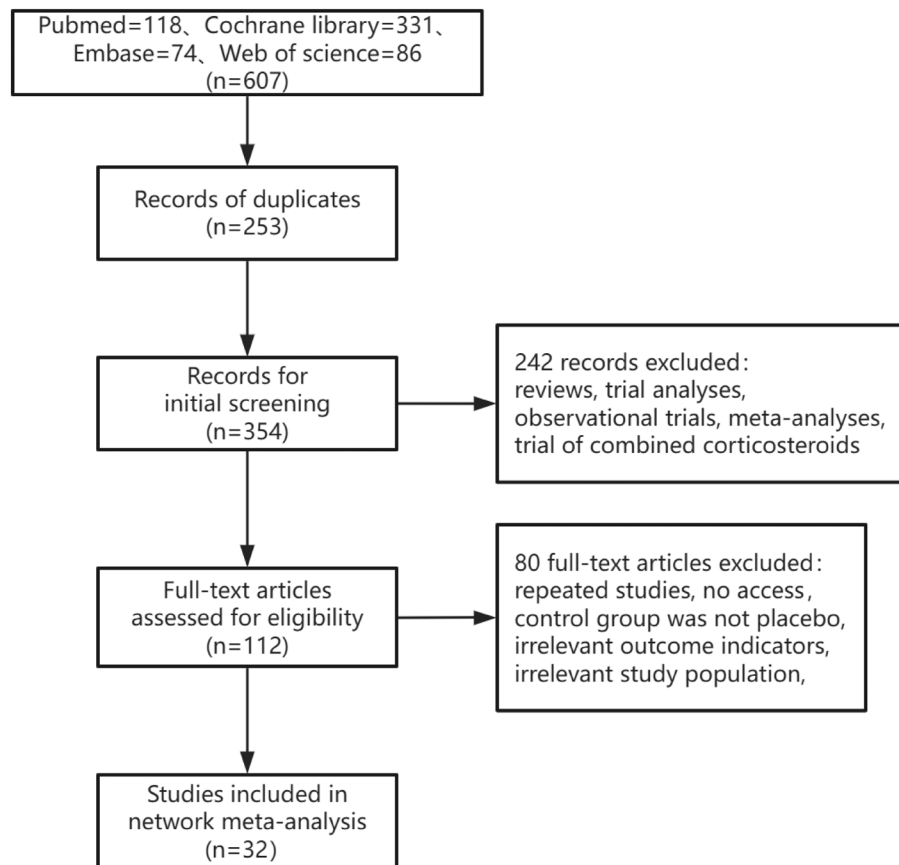


FIGURE 1 | Flow diagram of the selection process of literature.

of all literature, establishing and aligning screening criteria. A third reviewer examined texts presenting differing perspectives to determine whether or not to include them. We followed PRISMA guidelines for literature selection and presented a PRISMA flowchart. Data extraction utilized Excel 2019, with one individual inputting data and another verifying accuracy. The extracted content includes the name of the first author, the year of publication, study region, phase and trial identifier of the clinical trial, the number of patients, EASI score at baseline, age of patients, intervention measure, treatment duration (wk), outcome indicators.

2.4 | Statistical Analysis

We utilized RevMan 5.4 for evaluating literature quality and constructing the risk of bias maps. Network meta-analysis was performed using Stata 14 software. Data processing, generation of network evidence plots, calculation of surface under the cumulative ranking curve (SUCRA), league plots and funnel plots were conducted as part of the analysis. We used risk ratio (RR) and 95% confidence interval (95% CI) to analyze binary categorical variables, such as EASI-50, EASI-75, and EASI-90. Global inconsistency tests were performed. $p > 0.05$ indicates no significant inconsistency. SUCRA served as a visual tool to assess the impact of interventions on each outcome measure. Ranging from 0 to 100, SUCRA values indicate the effectiveness or safety of interventions, with values closer to 100 signifying higher effectiveness or safety [36]. A comparison-corrected funnel plot

was created to evaluate whether small-sample effects or publication bias were present.

3 | Results

3.1 | Literature Screening Results

In this study, a total of 607 articles were retrieved, of which 253 duplicates were excluded. After the initial screening, 354 articles were considered. Following a review of titles and abstracts, 242 articles were excluded, and 80 articles were excluded after full-text review. Ultimately, 32 articles were included. The literature search process was illustrated in Figure 1. The study encompassed 7588 patients, all diagnosed with moderate to severe AD. Among the interventions, Abrocitinib, Amltelimab, Astegolimab, CM310, Eblasakimab, Fezakinumab, Ustekinumab were each included by one study [32, 37–41]. Dupilumab was featured in 10 studies [19, 42–45], Lebrikizumab in 4 studies [22, 46, 47], Nemolizumab in 5 studies [25, 48–51], Tralokinumab in 3 studies [52, 53], and Rademikibart [54] and Spesolimab [55] in 2 studies, respectively. Further details were outlined in Table 1.

3.2 | Literature Quality Assessment

We evaluated the bias risk of the included literature according to the RCT bias assessment tool in Cochrane's manual. The bias risk

TABLE 1 | Study and patients' baseline characteristics of included RCTs.

	The name of the first author, the year of publication	Study region	Phase and trial identifier of clinical trial	The number of patients	EASI score at baseline, mean $\pm$ SD or range	Age of patients, mean $\pm$ SD or range	Intervention measure	Control group	Duration (wk)	Outcome indicators
1	Weidinger, 2023 [38]	19 sites in Germany, Poland, Spain and the UK	Phase II NCT03754309	78	32.9 $\pm$ 12.8	33.1 $\pm$ 15.0	Amltelimab 200 mg, q4w	Placebo	16	①②③④
					28.4 $\pm$ 11.5	35.6 $\pm$ 9.6	Amltelimab 500 mg, q4w			⑩⑪⑫
2	Maurer, 2022 [39]	21 sites in the United States, Poland, and Germany	Phase II NCT03747575	65	28.2 $\pm$ 8.0	39.7 $\pm$ 15.2	Astegolimab 490 mg, q4w	Placebo	16	②④⑤⑥⑦
										⑩⑪⑫
3	Zhao, 2024 [37]	21 sites in China	Phase II NCT04805411	120	26.3 $\pm$ 10.4	33.2 $\pm$ 13.2	CM310 300 mg, q2w	Placebo	16	①②③④
					30.1 $\pm$ 13.8	36.7 $\pm$ 16.4	CM310 150 mg, q2w			⑤⑥⑨
										⑩⑪⑫
4	Beck, 2014 [19]	13 sites in New South Wales Locations, Germany, New Zealand	Phase II NCT01385657	109	28.4 $\pm$ 1.8	33.7 $\pm$ 1.4	Dupilumab 300 mg, qw	Placebo	12	①②④⑤⑥⑧
										⑩⑪⑫
5	Simpson, 2016 [19]	103 sites in North America, Europe, and Asia	Phase III NCT02277743	671	30.4 (21.5–40.8)	38.0 (27.5–48.0)	Dupilumab 300 mg, q2w	Placebo	16	①②③④
					29.8 (22.0–41.2)	39.0 (27.0–51.0)	Dupilumab 300 mg, qw			⑤⑥⑦⑧⑨
										⑩⑪⑫
6	Simpson, 2016 [19]	103 sites in North America, Europe, and Asia	Phase III NCT02277769	708	28.6 (21.0–40.1)	34.0 (25.0–46.0)	Dupilumab 300 mg, q2w	Placebo	16	①②③④
					29.0 (21.2–41.8)	35.0 (25.0–46.0)	Dupilumab 300 mg, qw			⑤⑥⑦⑧⑨
										⑩⑪⑫
7	Thaci, 2016 [43]	91 sites in Canada, Czech Republic, Germany, Hungary, Japan, Poland, and the USA.	Phase II NCT01859988	379	30.1 $\pm$ 11.2	36.2 $\pm$ 10.7	Dupilumab 300 mg, qw	Placebo	16	①②③④
					33.8 $\pm$ 14.5	39.4 $\pm$ 12.1	Dupilumab 300 mg, q2w			⑤⑥⑦
					32.9 $\pm$ 15.5	35.8 $\pm$ 14.9	Dupilumab 200 mg, q2w			⑩⑪⑫
					29.4 $\pm$ 11.5	36.8 $\pm$ 10.8	Dupilumab 300 mg, q4w			
					32.2 $\pm$ 13.5	36.6 $\pm$ 11.6	Dupilumab 100 mg, q4w			

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TABLE 1 | (Continued)

	The name of the first author, the year of publication	Study region	Phase and trial identifier of clinical trial	The number of patients	EASI score at baseline, mean ± SD or range	Age of patients, mean ± SD or range	Intervention measure	Control group	Duration (wk)	Outcome indicators
8	Tsianakas, 2018 [44]	21 sites in Czechia, France, Germany, Hungary, Poland	Phase II NCT01548404	109	26.4 ± 2.4	37.3 ± 1.8	Dupilumab 300 mg, qw	Placebo	12	①④⑤⑥ ⑩⑪⑫
9	Guttman, 2019 [45]	5 sites in the United States and Canada	Phase II NCT01979016	54	33.4 ± 15.4	38.5 ± 13.6	Dupilumab 200 mg, qw	Placebo	16	①②③④ ⑤⑥⑦ ⑩⑪⑫
10	Simpson, 2020 [42]	45 sites in the United States and Canada	Phase III NCT03054428	169	35.8 ± 14.8	14.4 ± 1.6	Dupilumab 300 mg, q4w	Placebo	16	①②③④ ⑤⑥⑦⑧⑨ ⑩⑪⑫
11	Bieber, 2021 [56]	18 countries Australia and countries across North and South America, Europe, and Asia	Phase III NCT03720470	356	30.4 ± 12.0	37.1 ± 14.6	Dupilumab 300 mg, q2w	Placebo	16	①②③④⑥⑨ ⑩⑪⑫
12	Zhao, 2022 [57]	27 hospitals in China	Phase III NCT03912259	165	30.3 (22.8–42.9)	28.0 (24.0–35.0)	Dupilumab 300 mg, q2w	Placebo	16	①②③④ ⑤⑥⑧⑨ ⑩⑪⑫
13	Merola, 2023 [58]	42 sites in Australia, France, Germany, Israel, Italy, Spain, Switzerland, the United Arab Emirates, the UK and the USA	Phase III NCT04033367	157	26.2 ± 11.9	36.2 ± 14.7	Dupilumab 300 mg, q2w	Placebo	12	①②③⑤⑥⑨ ⑩⑪⑫
14	Veverka, 2024 [40]	10 sites in the United States, Australia, and Singapore	Phase II	40	32.9 ± 14.3 31.3 ± 12.3	32.5 ± 5.3 29.4 ± 4.9	Eblasakimab 200 mg, qw Eblasakimab 400 mg, qw	Placebo	8	①②③④ ⑩⑪⑫
15	Guttman, 2018 [32]	the USA	Phase II NCT01941537	60	30.5 ± 14.2 26.5 ± 5.5	34.0 ± 14.4 40.5 ± 14.9	Eblasakimab 600 mg, qw Fezakinumab 300 mg, q2w	Placebo	12	⑦⑩⑪⑫

(Continues)

TABLE 1 | (Continued)

	The name of the first author, the year of publication	Study region	Phase and trial identifier of clinical trial	The number of patients	EASI score at baseline, mean ± SD or range	Age of patients, mean ± SD or range	Intervention measure	Control group	Duration (wk)	Outcome indicators
16	Simpson, 2018 [46]	62 centers in the United States and Canada, Europe, and other countries	Phase II	104	26.9 ± 11.7	36.6 ± 12.3	Lebrikizumab 125 mg, q4w	Placebo	12	①②③④⑤⑨⑩⑪⑫
17	Guttman, 2020 [22]	57 centers in the USA	Phase II	280	29.9 ± 13.5	36.7 ± 16.5	Lebrikizumab 125 mg, q4w	Placebo	16	①②③④
			NCT03443024		26.2 ± 10.1	40.2 ± 17.9	Lebrikizumab 250 mg, q4w			⑤⑥⑦
					25.5 ± 11.2	38.9 ± 17.4	Lebrikizumab 250 mg, q2w			⑩⑪⑫
18	Silverberg, 2023 [47]	94 centers in the United States, Australia, Canada, Estonia, France, Korea, China, Latvia, Lithuania, Poland, Spain	Phase III	424	28.8 ± 11.3	36.1 ± 17.8	Lebrikizumab 250 mg, q2w	Placebo	16	②③④⑤⑥⑦⑧⑨⑩⑪⑫
			NCT04146363							
19	Silverberg, 2023 [47]	94 centers in the United States, Australia, Canada, Estonia, France, Korea, China, Latvia, Lithuania, Poland, Spain	Phase III	427	29.7 ± 12.0	36.6 ± 16.8	Lebrikizumab 250 mg, q2w	Placebo	16	②③④⑤⑥⑦⑧⑨⑩⑪⑫
			NCT04178967							
20	Mihara, 2019 [25]	18 centers in the USA	Phase II	184	35.0 ± 17.5	32.0 ± 10.9	Nemolizumab 0.1 mg/kg, q4w	Placebo	12	②③④⑤⑥⑦⑧⑨⑩⑪⑫
			NCT01986933		31.0 ± 18.5	35.0 ± 11.1	Nemolizumab 0.5 mg/kg, q4w			
					31.0 ± 12.6	38.0 ± 11.3	Nemolizumab 2.0 mg/kg, q4w			
21	Kabashima, 2020 [48]	Japan	Phase III	215	24.2 (10.0–65.0)	39.0 (13.0–73.0)	Nemolizumab 60 mg, q4w	Placebo	16	②③④⑤⑥⑦⑧⑨⑩⑪⑫

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TABLE 1 | (Continued)

	The name of the first author, the year of publication	Study region	Phase and trial identifier of clinical trial	The number of patients	EASI score at baseline, mean ± SD or range	Age of patients, mean ± SD or range	Intervention measure	Control group	Duration (wk)	Outcome indicators
22	Silverberg, 2020 [53]	58 sites in the United States, Canada, France, Germany, Poland, Australia	Phase II NCT03100344	226	24.8 ± 10.2	35.3 ± 14.8	Nemolizumab 10 mg, q4w Nemolizumab 30 mg, q4w Nemolizumab 90 mg, q4w	Placebo	24	①②③④ ⑤⑥⑦ ⑩⑪⑫
23	Silverberg, 2021 [50]	58 sites in the United States, Canada, France, Germany, Poland, Australia	Phase II	183	27.5 ± 10.2	40.1 ± 16.6	Nemolizumab 30 mg, q4w	Placebo	16	①②④⑩⑪⑫
24	Igarashi, 2023 [51]	Japan	Phase III	89	19.8 ± 7.9	9.0 ± 2.0	Nemolizumab 30 mg, q4w	Placebo	16	①②③④ ⑤⑥⑨ ⑩⑪⑫
25	Silverberg, 2024 [54]	59 sites in the United States, China, Australia, and New Zealand	Phase II NCT04444752	226	24.6 ± 10.5 27.6 ± 11.8 23.1 ± 8.2	39.5 ± 16.0 39.6 ± 14.8 41.7 ± 15.2	Rademikibart 150 mg, q2w Rademikibart 300 mg, q2w Rademikibart 300 mg, q4w	Placebo	16	④⑤⑧⑨ ⑩⑪⑫
26	Silverberg, 2024 [54]	China	Phase II NCT05017480	26	26.6 (16.5–38.5) 26.6 (16.0–54.0) 25.9 (16.3–45.5)	30.5 (23.0–70.0) 28.0 (21.0–41.0) 31.0 (23.0–59.0)	Rademikibart 150 mg, q2w Rademikibart 300 mg, q2w Rademikibart 300 mg, q4w	Placebo	16	①②③④⑤⑨ ⑩⑪⑫
27	Bissonnette, 2023 [55]	23 sites in Canada, Japan and the USA	Phase II NCT04086121	51	26.5 ± 10.1	43.2 ± 15.9	Spesolimab 600 mg, q4w	Placebo	16	②②④① ⑩⑪⑫
28	Bissonnette, 2023 [55]	23 sites in Canada, Japan and the USA	Phase II NCT03822832	40	26.5 ± 10.1	43.2 ± 15.9	Spesolimab 600 mg, q4w	Placebo	16	①②④⑤ ⑩⑪⑫

(Continues)

TABLE 1 | (Continued)

The name of the first author, the year of publication	Study region	Phase and trial identifier of clinical trial	The number of patients	EASI score at baseline, mean $\pm$ SD or range	Age of patients, mean $\pm$ SD or range	Intervention measure	Control group	Duration (wk)	Outcome indicators
29 Wollenberg, 2019 [53]	55 sites in Australia, Canada, Germany, Japan, Poland, and the United States	Phase II NCT02347176	204	24.8 $\pm$ 8.3	39.1 $\pm$ 15.1	Tralokinumab 45 mg, q2w Tralokinumab 150 mg, q2w Tralokinumab 300 mg, q2w	Placebo	12	①④⑩⑪⑫
30 Wollenberg, 2021 [52]	123 sites in the United States, France, Germany, Spain and Japan	Phase III NCT03131648	798	28.2 (21.3–40.0)	37.0 (27.0–48.0)	Tralokinumab 300 mg, q2w	Placebo	16	①②③④⑦⑨⑩
31 Wollenberg, 2021 [52]	123 sites in the United States, France, Germany, Spain and Japan	Phase III NCT03160885	792	28.2 (19.8–40.8)	34.0 (25.0–48.0)	Tralokinumab 300 mg, q2w	Placebo	16	①②③④⑦⑨⑩
32 Saeki, 2017 [41]	Japan	Phase II NCT01945086	79	38.0 $\pm$ 10.8 36.5 $\pm$ 12.6	38.5 (25.0–57.0) 34.0 (20.0–50.0)	Ustekinumab 45 mg, q4w Ustekinumab 90 mg, q4w	Placebo	12	①②④⑥⑨⑩

Note: ① Percentage of patients achieving 50% improvement in EASI score (EASI-50). ② Percentage of patients achieving 75% improvement in EASI score (EASI-75). ③ Percentage of patients achieving 90% improvement in EASI score (EASI-90). ④ Percentage of patients with an Investigator Global Assessment (IGA) Score of 0 or 1. ⑤ Percent change in Pruritus Numerical Rating Scale (NRS). ⑥ Percent change in EASI score. ⑦ Percent change in SCORing Atopic Dermatitis (SCORAD). ⑧ Change in Percent Body Surface Area (BSA). ⑨ Change in Dermatology Life Quality Index (DLQI). ⑩ Adverse events (AE). ⑪ Serious adverse events (SAE). ⑫ AE leading to discontinuation. Abbreviations: q2w: biweekly; q4w: once every 4 weeks; b.w: once a week; SD: standard deviation.

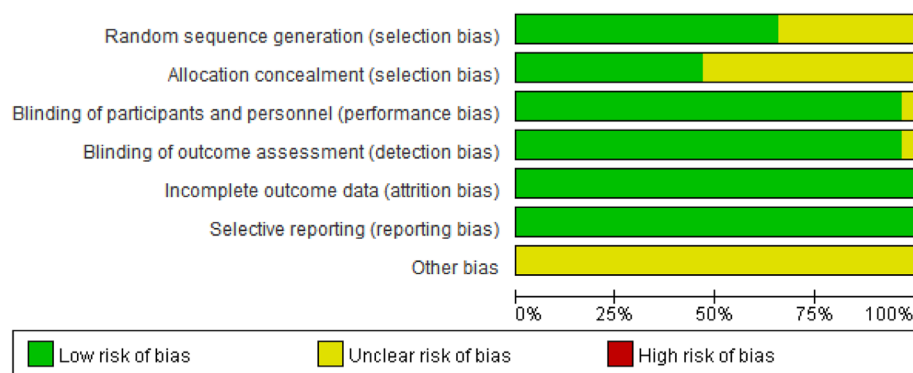


FIGURE 2 | Risk of bias graph for all included studies.

assessment comprehensively considered seven aspects, including random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias. As shown in Figures 2 and 3, all RCTs were labeled as “low risk” overall and the literature included in this study was of high quality.

3.3 | Global Inconsistency Tests

The global inconsistency test of primary efficacy indicators (percentage of patients in EASI-50, EASI-75, EASI-90, IGA score of 0 or 1) showed that p values were 0.7052, 0.7142, 0.8594 and 0.3520, respectively. Safety indicators were tested for global inconsistency, and the corresponding p values were 0.4449, 0.1969 and 0.3177. All p values were >0.05 , indicating that the global inconsistency was not significant.

3.4 | Network Evidence Plots

The results of network evidence plots for efficacy and safety indicators were shown in Figures 4 and 5. The primary efficacy indicators included 12 interventions. The safety indicators included 13 interventions. Each network diagram consisted of lines and dots. A straight line indicated direct comparative evidence between the two interventions. The thicker the lines in the diagram, the greater the number of studies that directly compared the two interventions. Using network meta-analysis, two unconnected interventions could be indirectly compared. The dots represented different interventions. The size of these points indicated the size of the total sample size for the intervention.

3.5 | Primary Efficacy Outcomes

3.5.1 | Percentage of Patients in EASI-50

A total of 26 studies were included, with a total of 7031 patients. The higher the SUCRA score, the more effective the intervention. There were 12 interventions involved, including placebo, Amltelimab, Astegolimab, CM310, Dupilumab, Eblasakimab, Lebrikizumab, Nemolizumab, Rademikibart, Spesolimab, Tralokinumab, Ustekinumab. The probability

ranking of interventions based on SUCRA values was shown in Table S1. The top three were Spesolimab (94.1%) $>$ Dupilumab (83.2%) $>$ Eblasakimab (75.9%). Cumulative probability graph results showed that Spesolimab had the greatest benefit for EASI-50, as shown in Figure S3a. The indicator EASI-50 was analyzed through league plots. Spesolimab was significantly more effective than placebo (RR = 5.45, 95% CI 1.31, 22.65). Dupilumab and Eblasakimab were also more effective than placebo (RR = 2.33, 95% CI 2.04, 2.67; RR = 2.28, 95% CI 1.23, 4.20). The above differences were statistically significant (Figure 6a).

3.5.2 | Percentage of Patients in EASI-75

A total of 26 studies, encompassing 7467 patients were analyzed. Table S1 illustrated the probability ranking of interventions based on SUCRA values. The top three interventions were Rademikibart (89.9%) $>$ Eblasakimab (77.9%) $>$ Dupilumab (77.8%). Cumulative probability graph results indicated that Rademikibart provided the greatest benefit for achieving EASI-75, as detailed in Figure S3b. Rademikibart exhibited significant efficacy over placebo (RR = 7.43, 95% CI 1.51–36.58) according to league plots analysis of EASI-75, as outlined in Figure 6b.

3.5.3 | Percentage of Patients in EASI-90

A total of 22 studies were included, with a total of 7109 patients. The probability ranking of interventions based on SUCRA values showed that the top three were Dupilumab (82.5%) $>$ CM310 (76.2%) $>$ Tralokinumab (63.1%) (Table S1). Cumulative probability graph results were shown in Figure S3c. Through league plots analysis of EASI-90, Dupilumab, CM310 and Tralokinumab were more effective than placebo, and the differences were statistically significant (RR = 4.52, 95% CI 3.72, 5.49; RR = 4.37, 95% CI 2.16, 8.82; RR = 3.43, 95% CI 2.17, 5.42) (Figure 6c).

3.5.4 | Percentage of Patients in IGA Score of 0 or 1

Twelve interventions were assessed in a comprehensive analysis comprising 28 studies and involving 8165 patients. Table S1 showed the probability ranking of interventions based on

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias)	Blinding of outcome assessment (detection bias)	Incomplete outcome data (attrition bias)	Selective reporting (reporting bias)	Other bias
Beck, 2014	+	?	+	+	+	+	?
Bieber, 2021	?	?	+	+	+	+	?
Bissonnette, 2023	?	?	+	+	+	+	?
Bissonnette, 2023*	?	?	?	?	+	+	?
Guttman Y, 2018	?	?	+	+	+	+	?
Guttman Y, 2019	?	?	+	+	+	+	?
Guttman Y, 2020	+	+	+	+	+	+	?
Igarashi, 2023	+	?	+	+	+	+	?
Kabashima, 2020	+	?	+	+	+	+	?
Maurer, 2022	+	+	+	+	+	+	?
Merola, 2023	+	+	+	+	+	+	?
Mihara, 2019	?	?	+	+	+	+	?
Saeki, 2017	+	+	+	+	+	+	?
Silverberg, 2020	?	?	+	+	+	+	?
Silverberg, 2021	?	?	+	+	+	+	?
Silverberg, 2023	+	+	+	+	+	+	?
Silverberg, 2023*	+	+	+	+	+	+	?
Silverberg, 2024	?	?	+	+	+	+	?
Simpson, 2016	+	+	+	+	+	+	?
Simpson, 2016*	+	+	+	+	+	+	?
Simpson, 2018	+	?	+	+	+	+	?
Simpson, 2020	+	+	+	+	+	+	?
Thaci, 2016	+	+	+	+	+	+	?
Tsianakas, 2018	+	+	+	+	+	+	?
Veverka, 2024	+	?	+	+	+	+	?
Wang, 2023	?	?	+	+	+	+	?
Weidinger, 2023	+	?	+	+	+	+	?
Wollenberg, 2019	?	?	+	+	+	+	?
Wollenberg, 2021	+	+	+	+	+	+	?
Wollenberg, 2021*	+	+	+	+	+	+	?
Zhao, 2024	+	+	+	+	+	+	?
Zhao, 2022	+	+	+	+	+	+	?

FIGURE 3 | Legend on next column.

FIGURE 3 | Risk of bias summary for all included studies. *Same author and same year but different studies to distinguish.

SUCRA values, with the top three being: Amltelimab (86.9%), Dupilumab (85.8%), and Spesolimab (69.7%). Cumulative probability graph result showed Amltelimab's superior efficacy, as evidenced by its highest SUCRA value (Figure S3d). In comparison with placebo, Amltelimab exhibited an RR of 4.88 and a 95% CI of (1.78, 13.36), as indicated in Figure 6d.

3.6 | Secondary Efficacy Outcomes

The efficacy was evaluated by the indicators of percent change in NRS, EASI and SCORAD. The results of the network meta-analysis all showed that Dupilumab had the highest SUCRA value (84.6%, 96.8%, 99.7%). In the analysis results of change in BSA and DLQI, the efficacy of Rademikibart was best (SUCRA = 91.9% and 100.0%) (Table S2, Figure S3).

3.7 | Safety Outcomes

Based on SUCRA values, the lowest incidences of AE, SAE and AE leading to discontinuation of the study were Astegolimab, Dupilumab and Tralokinumab, respectively (Table S4, Figure S5). Compared with Tralokinumab, placebo, Nemolizumab and Ustekinumab, Dupilumab had a lower incidence of SAE and the differences were statistically significant (RR=0.46, 95% CI 0.23–0.93; RR=0.43, 95% CI 0.30–0.63; RR=0.26, 95% CI 0.10–0.68; RR=0.26, 95% CI 0.11–0.62) (Figure 7). However, most of the differences among the treatment measures were not statistically significant.

3.8 | Publication Bias

Funnel plot results for efficacy and safety indicators showed that most scattering points were clustered at the top of the graph and distributed symmetrically (Figures S6 and S7). A small number of studies went beyond the funnel plot, suggesting possible publication bias or small sample effects in the included studies.

4 | Discussion

The study included 32 studies with a total of 7588 patients. The results of the pairwise meta-analysis found that Spesolimab had better efficacy in improving the EASI-50 compared to other interventions, with a SUCRA value of 94.1%. Rademikibart and Dupilumab had SUCRA values of 89.9% and 82.5%, respectively, and outperformed other interventions in improving EASI-75 and EASI-90. In addition, Amltelimab had a SUCRA value of 86.9%, which may have been the most favorable option for reducing IGA scores. Analysis of primary efficacy measures showed that all mAb were more effective than placebo. Dupilumab had better efficacy in improving NRS, EASI and SCORAD scores, with SUCRA values of 84.6%, 96.8% and 99.7%, respectively. Our findings also analyze the risk of adverse events associated with

these treatments. The use of the mAb group did not increase the statistical risk of AE and AE leading to discontinuation compared with the placebo group. Dupilumab has a better safety

profile than placebo, with a smaller SAE. The risk was statistically significant ($RR=0.43$, 95% CI 0.30–0.63). The SUCRA value was 84.2%.

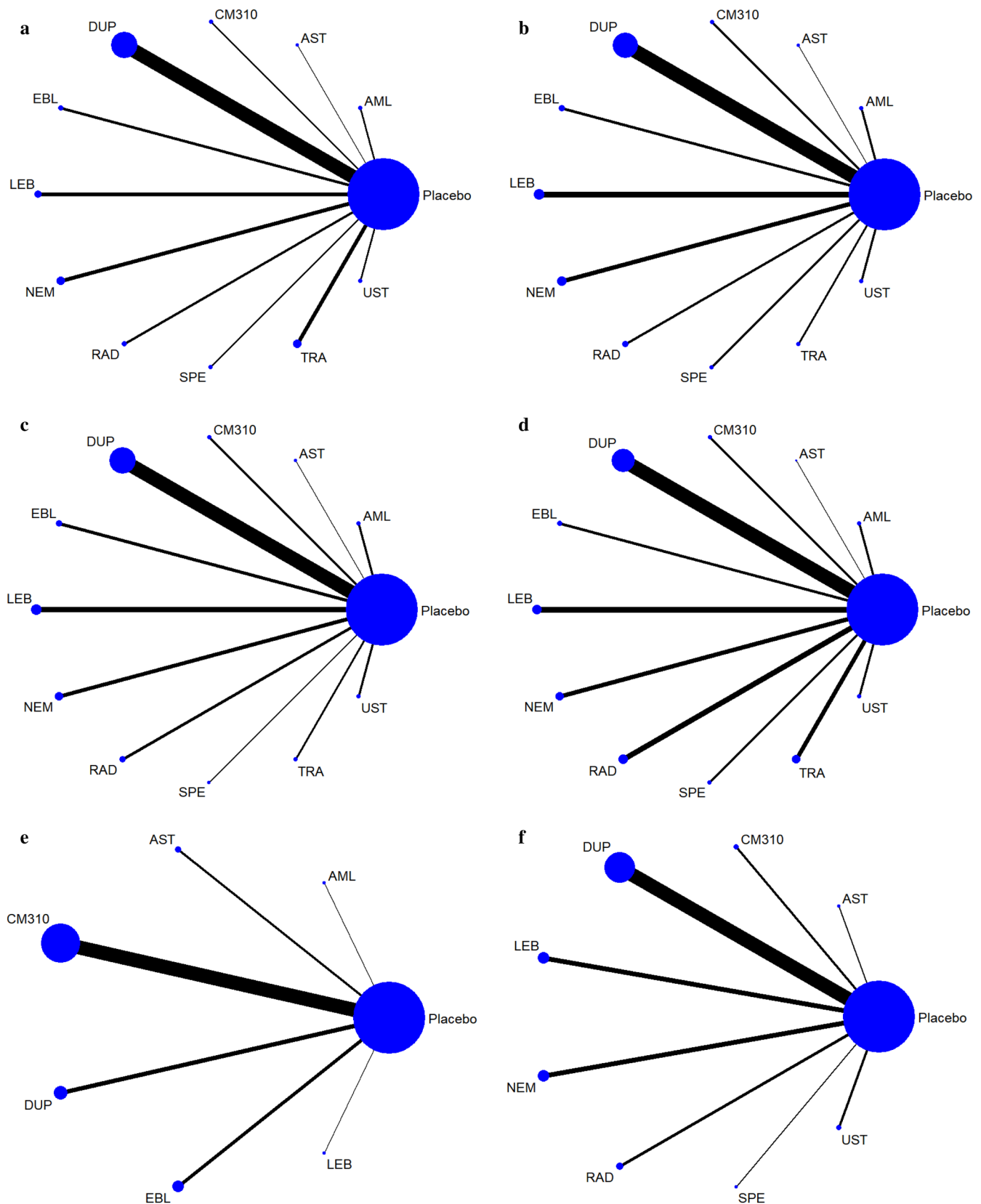


FIGURE 4 | Legend on next page.

FIGURE 4 | Network evidence plots of efficacy indicators. (a) Percentage of patients achieving 50% improvement in Eczema Area And Severity Index score (EASI-50), (b) Percentage of patients achieving 75% improvement in Eczema Area And Severity Index score (EASI-75), (c) Percentage of patients achieving 90% improvement in Eczema Area And Severity Index score (EASI-90), (d) Percentage of patients with an Investigator Global Assessment (IGA) Score of 0 or 1, (e) Percent change in Pruritus Numerical Rating Scale (NRS), (f) Percent change in EASI score, (g) Percent change in SCORing Atopic Dermatitis (SCORAD), (h) Change in Percent Body Surface Area (BSA), (i) Change in Dermatology Life Quality Index (DLQI). AML: Amltelimab, AST: Astegolimab, DUP: Dupilumab, EBL: Eblasakimab, FEZ: Fezakinumab, LEB: Lebrikizumab, NEM: Nemolizumab, RAD: Rademikibart, SPE: Spesolimab, TRA: Tralokinumab, UST: Ustekinumab.

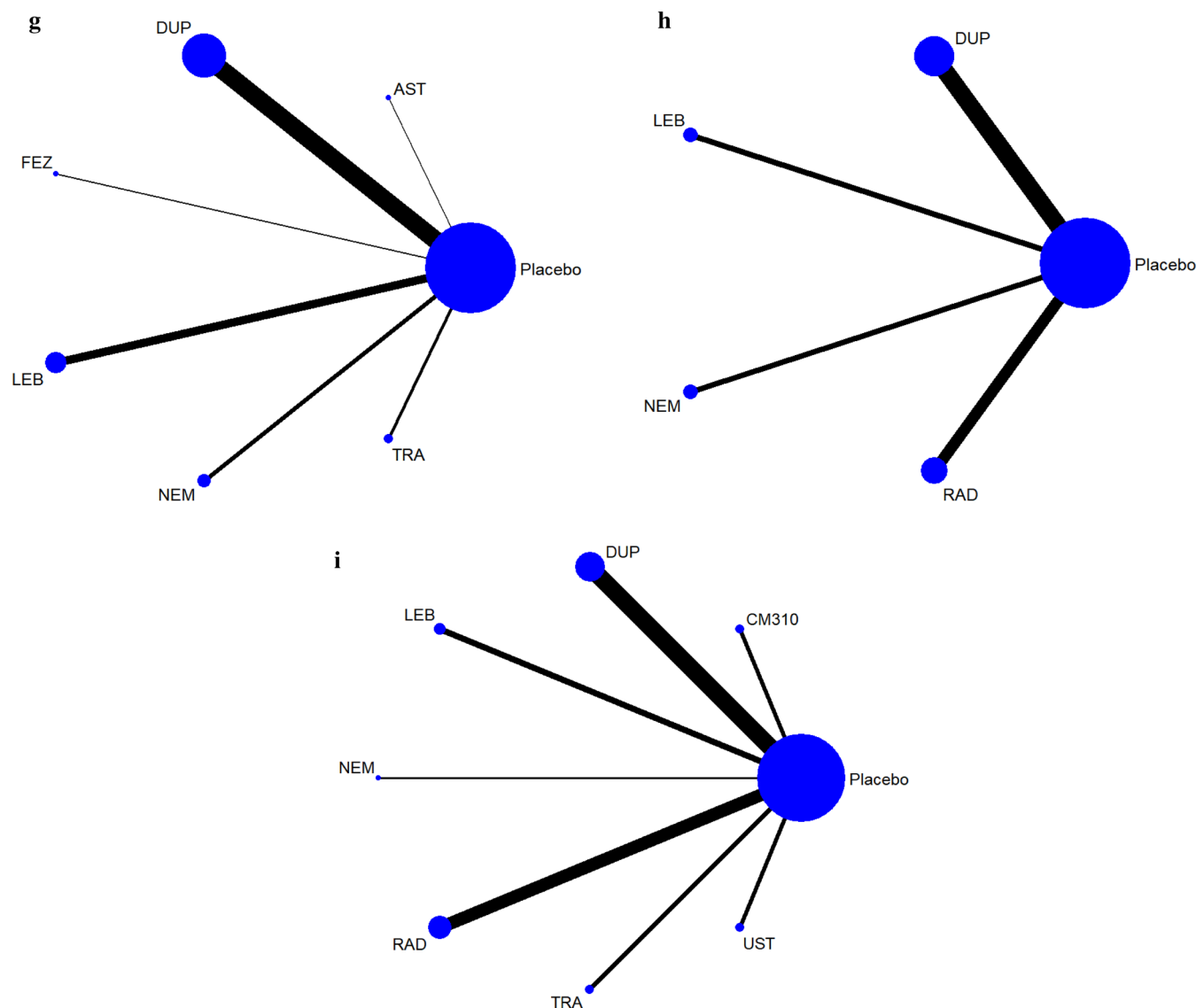


FIGURE 4 | (Continued)

IL-36 cytokines serve as the skin's frontline defense, tasked with detecting and responding to a multitude of external threats. They collaborate with other cytokines, chemokines, and immune-related molecules to modulate the host's defense mechanisms and regulate inflammatory pathways within the skin. Therefore, numerous studies have demonstrated the significant involvement of IL-36 cytokines in the pathogenesis of diverse skin disorders [59]. Dysregulation of IL-36 correlates with the onset of inflammatory conditions, with mounting evidence suggesting that it may play a key role in the pathological process of AE [60–64]. Furthermore, *Staphylococcus*

aureus infections associated with atopic dermatitis (AD) may exacerbate skin inflammation through IL-36-mediated T cell responses [65, 66]. Spesolimab, a humanized monoclonal immunoglobulin G antibody, treats patients with AD by obstructing the IL-36 pathway. During a Phase II clinical trial, patients administered with spesolimab exhibited enhancements in EASI scores compared to their initial baseline assessments [65]. IL-36 expression levels have been demonstrated to be elevated in lesional skin of patients with moderate to severe AD, and serum levels correlate positively with EASI scores in AD patients [60, 62].

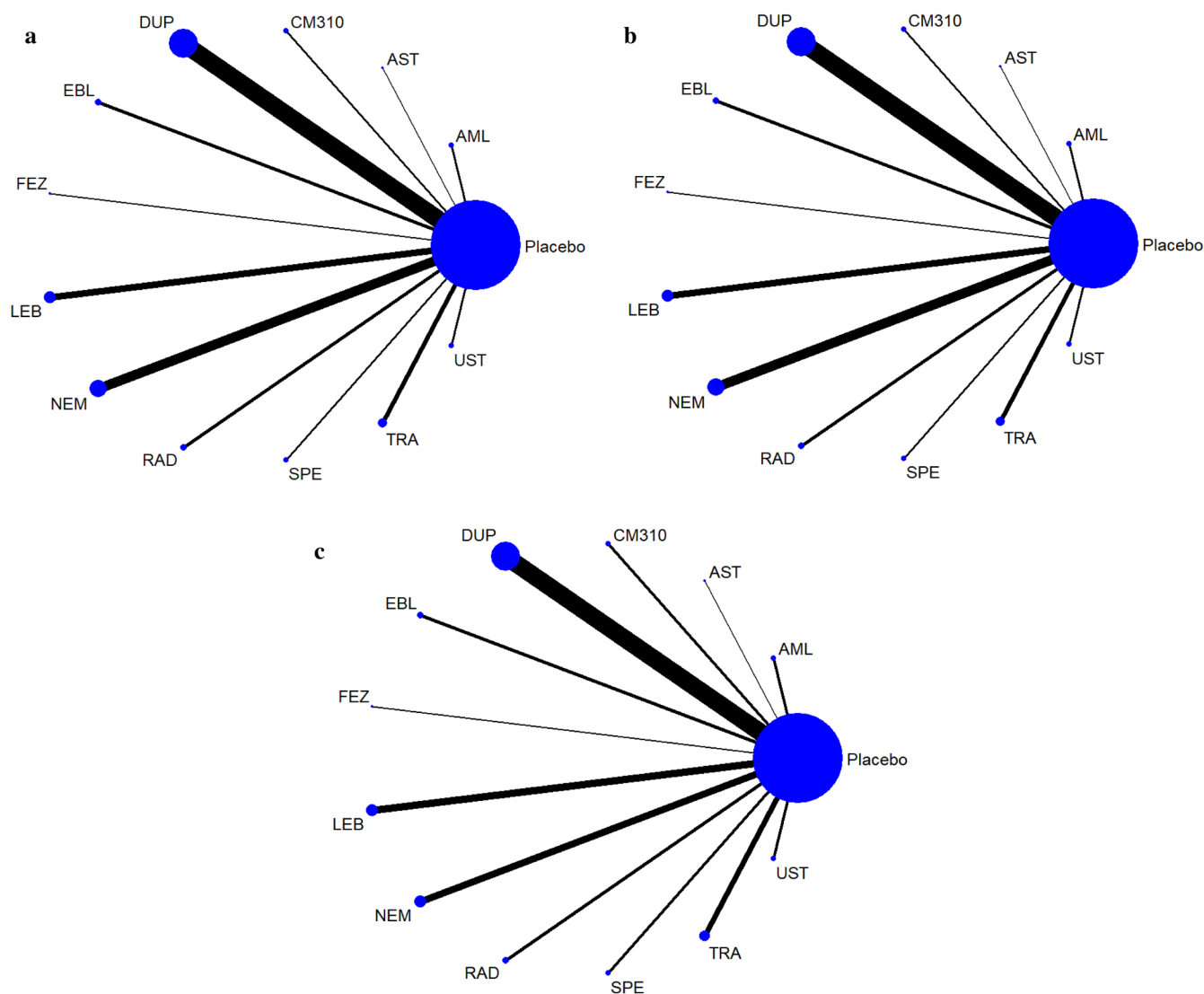


FIGURE 5 | Network evidence plots of safety indicators. (a) Adverse events (AE), (b) serious adverse events (SAE), (c) AE leading to discontinuation. AML: Amltelimab, AST: Astegolimab, DUP: Dupilumab, EBL: Eblasakimab, FEZ: Fezakinumab, LEB: Lebrikizumab, NEM: Nemolizumab, RAD: Rademikibart, SPE: Spesolimab, TRA: Tralokinumab, UST: Ustekinumab.

A monoclonal antibody targeting IL-4, Dupilumab inhibits receptor signaling in the Janus kinase-signal transducer and activator of transcription (JAK-STAT) pathway. This blockade interferes with three key mechanisms of AD: skin barrier dysfunction, Th2 cell differentiation, and IgE isotype switching [67]. A systematic review and meta-analysis of RCTs comparing Dupilumab with placebo demonstrated significant reductions in the EASI scores, percentage changes in SCORAD, and NRS scores, without an increase in AEs compared to placebo. Subgroup analyses confirmed these results in both adult and pediatric populations [68]. Furthermore, a comprehensive NMA assessing the effectiveness and safety of systemic AD treatments found that all active treatments, except Dupilumab, had a higher likelihood of therapeutic adverse events (TEAEs) compared to placebo [69]. Another meta-analysis of 50 RCTs provided robust, consistent, and high-quality evidence supporting the efficacy and safety of Dupilumab in adults over a one-year period [70]. Although Dupilumab is generally associated with a favorable safety profile, clinical trials and real-world applications have documented several AEs. Across 134 studies, 547 patients experienced AEs

during Dupilumab treatment. The most common AEs included facial and neck dermatitis, joint pain, alopecia, psoriasis, and cutaneous T-cell lymphoma. Most AEs resolved or improved upon discontinuation of Dupilumab, although three fatalities occurred due to serious adverse events (SAEs). Clinicians should remain vigilant for these AEs to ensure timely diagnosis and intervention [71]. As treatment duration increases, improvements in EASI scores suggest that Dupilumab may provide sustained long-term benefits for children and adolescents with AD [72]. A newer monoclonal antibody, Rademikibart, targets IL-4 receptor alpha (IL-4Ra). Preclinical studies have shown that Rademikibart has superior receptor binding affinity compared to Dupilumab [73]. Additionally, it demonstrated enhanced suppression of IL-4 and IL-13 in both transgenic mice and human skin samples [73]. Early clinical trials have confirmed its promising efficacy in AD patients, with fewer adverse effects and a lower incidence of conjunctivitis compared to Dupilumab [74]. Amltelimab, a novel therapeutic targeting OX40L, a key molecule in immune responses, has shown potential as a treatment for immune-mediated conditions [75]. In a Phase I trial involving

a

SPE	0.43 (0.10, 1.79)	0.42 (0.09, 1.97)	0.35 (0.07, 1.76)	0.30 (0.07, 1.35)	0.29 (0.07, 1.24)	0.29 (0.07, 1.25)	0.28 (0.07, 1.19)	0.27 (0.05, 1.33)	0.24 (0.06, 1.02)	0.16 (0.04, 0.76)	0.18 (0.04, 0.76)
2.34 (0.56, 9.77)	DUP	0.98 (0.52, 1.83)	0.82 (0.38, 1.77)	0.70 (0.43, 1.15)	0.69 (0.53, 0.90)	0.67 (0.46, 0.99)	0.65 (0.49, 0.87)	0.62 (0.29, 1.34)	0.56 (0.43, 0.74)	0.38 (0.22, 0.68)	0.43 (0.37, 0.49)
2.40 (0.51, 11.30)	1.03 (0.55, 1.92)	EBL	0.84 (0.32, 2.23)	0.72 (0.33, 1.57)	0.70 (0.37, 1.36)	0.69 (0.34, 1.41)	0.67 (0.35, 1.30)	0.64 (0.24, 1.68)	0.58 (0.30, 1.12)	0.39 (0.17, 0.90)	0.44 (0.24, 0.81)
2.86 (0.57, 14.39)	1.22 (0.56, 2.65)	1.19 (0.45, 3.17)	RAD	0.86 (0.35, 2.12)	0.84 (0.38, 1.86)	0.82 (0.35, 1.92)	0.80 (0.36, 1.79)	0.76 (0.26, 2.22)	0.69 (0.31, 1.53)	0.47 (0.18, 1.21)	0.52 (0.24, 1.12)
3.32 (0.74, 14.87)	1.42 (0.87, 2.32)	1.38 (0.64, 3.00)	1.16 (0.47, 2.84)	AML	0.97 (0.58, 1.65)	0.96 (0.53, 1.74)	0.93 (0.54, 1.59)	0.88 (0.36, 2.15)	0.80 (0.47, 1.36)	0.55 (0.26, 1.13)	0.61 (0.38, 0.98)
3.40 (0.80, 14.39)	1.46 (1.12, 1.90)	1.42 (0.74, 2.73)	1.19 (0.54, 2.63)	1.03 (0.61, 1.73)	TRA	0.98 (0.64, 1.51)	0.95 (0.68, 1.34)	0.91 (0.41, 1.99)	0.82 (0.59, 1.14)	0.56 (0.31, 1.02)	0.62 (0.50, 0.78)
3.47 (0.80, 15.09)	1.48 (1.01, 2.19)	1.45 (0.71, 2.96)	1.21 (0.52, 2.82)	1.05 (0.58, 1.90)	1.02 (0.66, 1.57)	CM310	0.97 (0.63, 1.51)	0.93 (0.40, 2.13)	0.84 (0.54, 1.29)	0.57 (0.29, 1.11)	0.64 (0.44, 0.92)
3.57 (0.84, 15.15)	1.53 (1.15, 2.03)	1.49 (0.77, 2.89)	1.25 (0.56, 2.78)	1.08 (0.63, 1.84)	1.05 (0.75, 1.47)	1.03 (0.66, 1.60)	LEB	0.95 (0.43, 2.10)	0.86 (0.61, 1.22)	0.59 (0.32, 1.08)	0.65 (0.51, 0.84)
3.75 (0.75, 18.74)	1.60 (0.75, 3.44)	1.56 (0.59, 4.12)	1.31 (0.45, 3.82)	1.13 (0.47, 2.74)	1.10 (0.50, 2.41)	1.08 (0.47, 2.49)	1.05 (0.48, 2.32)	AST	0.90 (0.41, 1.99)	0.62 (0.24, 1.57)	0.69 (0.32, 1.46)
4.15 (0.98, 17.57)	1.77 (1.35, 2.33)	1.73 (0.90, 3.34)	1.45 (0.65, 3.22)	1.25 (0.74, 2.12)	1.22 (0.88, 1.70)	1.19 (0.77, 1.85)	1.16 (0.82, 1.64)	1.11 (0.50, 2.43)	NEM	0.68 (0.37, 1.25)	0.76 (0.60, 0.97)
6.07 (1.32, 28.03)	2.60 (1.46, 4.61)	2.53 (1.11, 5.81)	2.12 (0.83, 5.46)	1.83 (0.88, 3.81)	1.78 (0.98, 3.26)	1.75 (0.90, 3.41)	1.70 (0.92, 3.14)	1.62 (0.64, 4.12)	1.46 (0.80, 2.69)	UST	1.11 (0.64, 1.95)
5.45 (1.31, 22.65)	2.33 (2.04, 2.67)	2.28 (1.23, 4.20)	1.91 (0.89, 4.08)	1.64 (1.02, 2.64)	1.60 (1.28, 2.01)	1.57 (1.09, 2.26)	1.53 (1.19, 1.96)	1.45 (0.69, 3.08)	1.31 (1.04, 1.67)	0.90 (0.51, 1.57)	Placebo

FIGURE 6 | League plots of EASI-50 (a), EASI-75 (b), EASI-90 (c), IGA (d). Bold texts indicated that the comparison between the two groups was statistically significant. AML: Amltelimab, AST: Astegolimab, DUP: Dupilumab, EBL: Eblasakimab, FEZ: Fezakinumab, LEB: Lebrikizumab, NEM: Nemolizumab, RAD: Rademikibart, SPE: Spesolimab, TRA: Tralokinumab, UST: Ustekinumab. EASI-50: Percentage of patients achieving 50% improvement in Eczema Area And Severity Index score; EASI-75: Percentage of patients achieving 75% improvement in Eczema Area And Severity Index score; EASI-90: Percentage of patients achieving 90% improvement in Eczema Area And Severity Index score; IGA: Percentage of patients with an Investigator Global Assessment Score of 0 or 1.

healthy volunteers, pharmacodynamic results suggested that Amltelimab could be a promising therapeutic option [28, 76]. In a Phase II clinical trial, Amltelimab demonstrated promising efficacy and an acceptable safety profile in patients with moderate to severe AD, particularly in those for whom local therapies were insufficient [38]. The drug modulates co-stimulatory signals during antigen presentation, which may enhance disease control. Monoclonal antibody therapies represent a promising and effective approach for moderate to severe AD. Ongoing research aims to further modulate the molecular and cellular

mechanisms implicated in AD pathophysiology, with the goal of expanding and improving available treatment options.

The safety of these treatments is a crucial factor in guiding therapeutic decisions, especially for long-term management. The monoclonal antibody treatments, particularly dupilumab, showed a favorable safety profile, with fewer injection-site reactions and systemic adverse events compared to placebo [67]. The most common AEs in the monoclonal antibody group were mild to moderate in nature, including mild upper respiratory tract

b

RAD	0.52 (0.09, 3.17)	0.47 (0.09, 2.31)	0.45 (0.08, 2.43)	0.37 (0.04, 3.24)	0.33 (0.06, 1.63)	0.32 (0.06, 1.65)	0.30 (0.05, 1.64)	0.20 (0.03, 1.27)	0.21 (0.04, 1.08)	0.12 (0.02, 0.80)	0.13 (0.03, 0.66)
1.93 (0.32, 11.76)	EBL	0.90 (0.38, 2.14)	0.88 (0.32, 2.38)	0.71 (0.13, 3.92)	0.63 (0.26, 1.52)	0.62 (0.24, 1.57)	0.58 (0.20, 1.63)	0.38 (0.10, 1.38)	0.41 (0.17, 1.02)	0.24 (0.06, 0.88)	0.26 (0.11, 0.61)
2.15 (0.43, 10.68)	1.12 (0.47, 2.66)	DUP	0.98 (0.56, 1.69)	0.79 (0.18, 3.51)	0.70 (0.52, 0.93)	0.69 (0.46, 1.03)	0.64 (0.35, 1.19)	0.42 (0.16, 1.13)	0.46 (0.33, 0.65)	0.26 (0.10, 0.72)	0.29 (0.25, 0.34)
2.20 (0.41, 11.79)	1.14 (0.42, 3.11)	1.02 (0.59, 1.77)	CM310	0.81 (0.17, 3.89)	0.72 (0.40, 1.27)	0.71 (0.37, 1.34)	0.66 (0.30, 1.46)	0.43 (0.14, 1.30)	0.47 (0.26, 0.86)	0.27 (0.09, 0.83)	0.30 (0.18, 0.50)
2.72 (0.31, 24.02)	1.41 (0.26, 7.82)	1.27 (0.28, 5.63)	1.24 (0.26, 5.96)	SPE	0.89 (0.20, 3.97)	0.87 (0.19, 4.03)	0.81 (0.16, 4.03)	0.53 (0.09, 3.14)	0.58 (0.13, 2.64)	0.33 (0.06, 1.99)	0.37 (0.08, 1.62)
3.08 (0.61, 15.41)	1.60 (0.66, 3.86)	1.43 (1.07, 1.91)	1.40 (0.79, 2.48)	1.13 (0.25, 5.07)	LEB	0.99 (0.64, 1.53)	0.92 (0.48, 1.74)	0.60 (0.22, 1.64)	0.66 (0.45, 0.96)	0.38 (0.14, 1.05)	0.41 (0.33, 0.52)
3.12 (0.61, 16.03)	1.62 (0.64, 4.10)	1.45 (0.97, 2.18)	1.42 (0.75, 2.69)	1.15 (0.25, 5.28)	1.01 (0.65, 1.57)	TRA	0.93 (0.46, 1.88)	0.61 (0.22, 1.73)	0.66 (0.41, 1.07)	0.38 (0.13, 1.11)	0.42 (0.29, 0.61)
3.34 (0.61, 18.35)	1.74 (0.61, 4.91)	1.56 (0.84, 2.89)	1.52 (0.69, 3.36)	1.23 (0.25, 6.07)	1.09 (0.57, 2.06)	1.07 (0.53, 2.16)	AML	0.65 (0.21, 2.05)	0.71 (0.37, 1.39)	0.41 (0.13, 1.31)	0.45 (0.25, 0.82)
5.11 (0.79, 33.07)	2.65 (0.73, 9.67)	2.37 (0.88, 6.37)	2.32 (0.77, 7.01)	1.88 (0.32, 11.05)	1.66 (0.61, 4.52)	1.64 (0.58, 4.64)	1.53 (0.49, 4.78)	AST	1.09 (0.39, 3.02)	0.63 (0.16, 2.52)	0.69 (0.26, 1.82)
4.69 (0.93, 23.78)	2.43 (0.98, 6.02)	2.18 (1.55, 3.07)	2.13 (1.16, 3.90)	1.72 (0.38, 7.83)	1.53 (1.04, 2.24)	1.50 (0.93, 2.43)	1.40 (0.72, 2.74)	0.92 (0.33, 2.55)	NEM	0.58 (0.20, 1.63)	0.63 (0.47, 0.86)
8.15 (1.24, 53.41)	4.23 (1.14, 15.69)	3.79 (1.38, 10.40)	3.70 (1.20, 11.40)	2.99 (0.50, 17.86)	2.65 (0.95, 7.37)	2.61 (0.90, 7.56)	2.44 (0.76, 7.78)	1.60 (0.40, 6.43)	1.74 (0.61, 4.92)	UST	1.10 (0.41, 2.97)
7.43 (1.51, 36.58)	3.85 (1.64, 9.04)	3.45 (2.93, 4.07)	3.37 (2.00, 5.69)	2.73 (0.62, 12.01)	2.41 (1.91, 3.05)	2.38 (1.64, 3.45)	2.22 (1.22, 4.03)	1.45 (0.55, 3.85)	1.58 (1.17, 2.14)	0.91 (0.34, 2.46)	Placebo

FIGURE 6 | (Continued)

infections and local reactions at the injection site. These adverse events were generally transient and manageable. Importantly, dupilumab was associated with a lower incidence of SAEs, such as serious infections and anaphylactic reactions, which are often concerns with biologic therapies [41]. Given that these therapies may require long-term use, the reduced incidence of severe adverse events with dupilumab, as evidenced by the statistical significance ($RR=0.43$), provides reassurance regarding its long-term safety. Clinicians should consider both the efficacy and safety profiles when making treatment decisions, especially for patients who are at risk of experiencing side effects with conventional therapies or have a history of severe AEs. While dupilumab demonstrated a superior safety profile with fewer SAEs, other mAb showed comparable safety outcomes to placebo, with no significant increase in the risk of AEs or SAEs. This suggests

that while all the monoclonal antibody treatments assessed in this study are generally well-tolerated, dupilumab may offer an advantage for patients requiring long-term therapy, especially for those with previous concerns about biologic treatments [76]. Although efficacy remains a primary concern in selecting a treatment for atopic dermatitis, the safety profile is equally critical, particularly for patients who may require long-term or repeated use of biologic therapies. The favorable safety data for monoclonal antibodies, especially dupilumab, suggest that these treatments may be a suitable option for patients who have failed traditional therapies or for those who are concerned about the side effects of systemic treatments.

Although there was extensive research on the use of monoclonal antibodies in the treatment of AD, our study presented a unique

c

DUP	0.97 (0.47, 2.01)	0.76 (0.46, 1.25)	0.75 (0.53, 1.07)	0.78 (0.15, 4.23)	0.62 (0.03, 12.31)	0.56 (0.23, 1.33)	0.54 (0.11, 2.60)	0.55 (0.21, 1.46)	0.44 (0.27, 0.72)	0.24 (0.04, 1.58)	0.22 (0.18, 0.27)
1.03 (0.50, 2.15)	CM310	0.79 (0.34, 1.82)	0.77 (0.36, 1.66)	0.81 (0.13, 4.99)	0.64 (0.03, 13.73)	0.58 (0.19, 1.73)	0.56 (0.10, 3.09)	0.57 (0.18, 1.87)	0.45 (0.20, 1.05)	0.25 (0.03, 1.84)	0.23 (0.11, 0.46)
1.32 (0.80, 2.16)	1.27 (0.55, 2.94)	TRA	0.99 (0.57, 1.70)	1.03 (0.18, 5.86)	0.81 (0.04, 16.66)	0.73 (0.28, 1.92)	0.71 (0.14, 3.61)	0.73 (0.25, 2.09)	0.58 (0.30, 1.11)	0.31 (0.05, 2.17)	0.29 (0.18, 0.46)
1.34 (0.94, 1.90)	1.29 (0.60, 2.77)	1.01 (0.59, 1.75)	LEB	1.05 (0.19, 5.73)	0.83 (0.04, 16.57)	0.74 (0.30, 1.82)	0.72 (0.15, 3.53)	0.74 (0.27, 2.00)	0.59 (0.34, 1.01)	0.32 (0.05, 2.13)	0.30 (0.22, 0.40)
1.28 (0.24, 6.89)	1.23 (0.20, 7.58)	0.97 (0.17, 5.50)	0.95 (0.17, 5.23)	RAD	0.79 (0.03, 24.16)	0.71 (0.11, 4.64)	0.68 (0.07, 6.78)	0.71 (0.10, 4.84)	0.56 (0.10, 3.18)	0.30 (0.02, 3.77)	0.28 (0.05, 1.51)
1.62 (0.08, 32.16)	1.56 (0.07, 33.52)	1.23 (0.06, 25.14)	1.21 (0.06, 24.27)	1.27 (0.04, 38.82)	SPE	0.90 (0.04, 20.03)	0.87 (0.03, 25.23)	0.89 (0.04, 20.48)	0.71 (0.03, 14.55)	0.39 (0.01, 13.10)	0.36 (0.02, 7.07)
1.79 (0.75, 4.28)	1.73 (0.58, 5.21)	1.36 (0.52, 3.57)	1.34 (0.55, 3.29)	1.41 (0.22, 9.19)	1.11 (0.05, 24.68)	AML	0.96 (0.16, 5.71)	0.99 (0.28, 3.54)	0.79 (0.30, 2.07)	0.43 (0.05, 3.36)	0.40 (0.17, 0.93)
1.86 (0.38, 9.03)	1.80 (0.32, 10.02)	1.42 (0.28, 7.24)	1.40 (0.28, 6.86)	1.46 (0.15, 14.47)	1.15 (0.04, 33.52)	1.04 (0.18, 6.16)	AST	1.03 (0.16, 6.43)	0.82 (0.16, 4.19)	0.44 (0.04, 5.13)	0.41 (0.09, 1.97)
1.81 (0.69, 4.77)	1.75 (0.54, 5.70)	1.37 (0.48, 3.94)	1.35 (0.50, 3.66)	1.42 (0.21, 9.73)	1.12 (0.05, 25.63)	1.01 (0.28, 3.60)	0.97 (0.16, 6.06)	EBL	0.80 (0.28, 2.28)	0.43 (0.05, 3.54)	0.40 (0.15, 1.04)
2.27 (1.38, 3.74)	2.20 (0.95, 5.09)	1.73 (0.90, 3.30)	1.70 (0.99, 2.94)	1.78 (0.31, 10.12)	1.41 (0.07, 28.81)	1.27 (0.48, 3.32)	1.22 (0.24, 6.24)	1.26 (0.44, 3.61)	NEM	0.54 (0.08, 3.75)	0.50 (0.32, 0.80)
4.20 (0.63, 27.79)	4.06 (0.54, 30.20)	3.19 (0.46, 22.08)	3.14 (0.47, 21.07)	3.29 (0.27, 40.81)	2.60 (0.08, 88.32)	2.34 (0.30, 18.38)	2.25 (0.19, 26.02)	2.32 (0.28, 19.08)	1.85 (0.27, 12.78)	UST	0.93 (0.14, 6.09)
4.52 (3.72, 5.49)	4.37 (2.16, 8.82)	3.43 (2.17, 5.42)	3.38 (2.52, 4.54)	3.54 (0.66, 18.90)	2.79 (0.14, 55.23)	2.52 (1.08, 5.86)	2.42 (0.51, 11.61)	2.50 (0.97, 6.46)	1.99 (1.26, 3.14)	1.08 (0.16, 7.06)	Placebo

FIGURE 6 | (Continued)

contribution in several key aspects: (1)Scope of trial groups: Existing meta-analyses predominantly focus on specific monoclonal antibodies or comparisons between monoclonal antibodies and JAK inhibitors [77, 78]. In contrast, our study included all monoclonal antibodies that have received clinical approval, offering a more comprehensive evaluation. (2) Type of analysis: While most studies to date were meta-analyses, our research was the first to employ an NMA approach to compare all approved monoclonal antibodies, providing a broader comparative perspective. (3)Timeliness of data: Our study included data up to November 2024, making it more current compared to existing research and reflecting the latest advancements in the field. In addition, all RCTs in our study were considered to be of high

quality. Efficacy was ranked by SUCRA value to improve the reliability of statistical results and more accurate data.

However, there were some limitations to our study. Firstly, meta-analysis data only came from published literature, and some negative results and non-statistical data could not be published. The number of studies on relevant interventions included varied, such as Dupilumab, which included 10 studies, while Amlitelimab included only one study. The number of included studies was limited and, therefore, there was publication bias. Secondly, most studies came from placebo-controlled RCTs, and the number of head-to-head studies of monoclonal antibodies was extremely limited, leading to indirect estimates

d

AML	0.82 (0.29, 2.28)	0.80 (0.08, 7.98)	0.51 (0.16, 1.60)	0.50 (0.08, 3.25)	0.46 (0.16, 1.31)	0.43 (0.13, 1.47)	0.40 (0.14, 1.15)	0.35 (0.08, 1.56)	0.32 (0.11, 0.94)	0.21 (0.06, 0.71)	0.21 (0.07, 0.56)
1.22 (0.44, 3.40)	DUP	0.98 (0.12, 7.78)	0.62 (0.34, 1.10)	0.61 (0.12, 2.99)	0.56 (0.40, 0.79)	0.52 (0.25, 1.09)	0.49 (0.34, 0.71)	0.43 (0.14, 1.30)	0.39 (0.26, 0.59)	0.26 (0.13, 0.52)	0.25 (0.21, 0.30)
1.25 (0.13, 12.40)	1.02 (0.13, 8.12)	SPE	0.63 (0.07, 5.35)	0.62 (0.05, 8.37)	0.57 (0.07, 4.61)	0.54 (0.06, 4.75)	0.50 (0.06, 4.03)	0.44 (0.04, 4.52)	0.40 (0.05, 3.27)	0.27 (0.03, 2.33)	0.26 (0.03, 2.02)
1.98 (0.63, 6.23)	1.62 (0.91, 2.90)	1.59 (0.19, 13.44)	RAD	0.98 (0.18, 5.26)	0.91 (0.49, 1.69)	0.85 (0.35, 2.09)	0.79 (0.42, 1.49)	0.69 (0.20, 2.37)	0.64 (0.33, 1.23)	0.42 (0.18, 1.00)	0.41 (0.23, 0.70)
2.01 (0.31, 13.16)	1.65 (0.33, 8.14)	1.61 (0.12, 21.79)	1.02 (0.19, 5.45)	AST	0.93 (0.19, 4.63)	0.86 (0.15, 4.91)	0.81 (0.16, 4.06)	0.70 (0.10, 4.85)	0.65 (0.13, 3.30)	0.43 (0.08, 2.39)	0.41 (0.08, 2.01)
2.17 (0.76, 6.18)	1.78 (1.27, 2.51)	1.74 (0.22, 14.00)	1.10 (0.59, 2.04)	1.08 (0.22, 5.40)	LEB	0.93 (0.44, 2.00)	0.87 (0.58, 1.32)	0.76 (0.24, 2.36)	0.70 (0.44, 1.11)	0.46 (0.23, 0.95)	0.45 (0.34, 0.59)
2.33 (0.68, 7.99)	1.91 (0.92, 3.98)	1.87 (0.21, 16.58)	1.18 (0.48, 2.90)	1.16 (0.20, 6.58)	1.07 (0.50, 2.29)	CM310	0.93 (0.43, 2.03)	0.81 (0.22, 3.02)	0.75 (0.34, 1.67)	0.50 (0.19, 1.31)	0.48 (0.23, 0.97)
2.50 (0.87, 7.18)	2.05 (1.41, 2.96)	2.00 (0.25, 16.17)	1.26 (0.67, 2.39)	1.24 (0.25, 6.25)	1.15 (0.76, 1.74)	1.07 (0.49, 2.33)	TRA	0.87 (0.28, 2.74)	0.80 (0.50, 1.31)	0.53 (0.26, 1.11)	0.51 (0.37, 0.70)
2.86 (0.64, 12.74)	2.35 (0.77, 7.18)	2.30 (0.22, 23.85)	1.45 (0.42, 4.97)	1.42 (0.21, 9.81)	1.32 (0.42, 4.10)	1.23 (0.33, 4.56)	1.15 (0.36, 3.61)	EBL	0.92 (0.29, 2.95)	0.61 (0.17, 2.21)	0.59 (0.19, 1.77)
3.10 (1.06, 9.07)	2.55 (1.69, 3.83)	2.49 (0.31, 20.28)	1.57 (0.81, 3.04)	1.54 (0.30, 7.85)	1.43 (0.90, 2.27)	1.33 (0.60, 2.96)	1.24 (0.77, 2.02)	1.08 (0.34, 3.47)	NEM	0.66 (0.31, 1.41)	0.64 (0.44, 0.92)
4.68 (1.40, 15.58)	3.83 (1.93, 7.60)	3.75 (0.43, 32.77)	2.37 (1.00, 5.59)	2.33 (0.42, 12.94)	2.15 (1.05, 4.40)	2.01 (0.76, 5.29)	1.87 (0.90, 3.89)	1.63 (0.45, 5.90)	1.51 (0.71, 3.20)	UST	0.96 (0.50, 1.85)
4.88 (1.78, 13.36)	4.00 (3.32, 4.82)	3.91 (0.50, 30.84)	2.47 (1.42, 4.28)	2.42 (0.50, 11.83)	2.24 (1.70, 2.96)	2.09 (1.03, 4.26)	1.95 (1.42, 2.68)	1.70 (0.57, 5.13)	1.57 (1.09, 2.26)	1.04 (0.54, 2.01)	Placebo

FIGURE 6 | (Continued)

of all comparisons in our NMA and affecting the accuracy of the results. Future RCTs should focus more on direct comparisons between monoclonal antibodies. Thirdly, we did not perform subgroup analyses for different doses and duration of treatment, which may have biased the results. Fourthly, due to article length limitations, some other efficacy endpoints were not included in the analysis, such as the percentage of patients with ≥ 2 -point improvement in IGA from baseline, and the percentage of patients with ≥ 3 or ≥ 4 -point reduction in NRS score. Finally, the main analysis period of this study was 8–24 weeks; long-term safety data and real-world data from patients may be more clinically significant. This will continue to be determined in future studies. Based on the NMA of monoclonal antibodies

in existing studies, clinical benefits have been observed in the treatment of AD, but whether it is really worthy of clinical promotion remains to be further discussed. The limitations in the safety analysis of this study suggest that future research should include longer follow-up periods and larger sample sizes to obtain more comprehensive safety data. This is particularly pertinent for assessing the risks of monoclonal antibodies that did not demonstrate significant differences. In subsequent studies, we will incorporate additional clinical data to discuss the clinical relevance of these safety outcomes.

NMA provided effective evidence for evaluating the efficacy and safety of monoclonal antibody in the treatment of AD.

a

AST	1.13 (0.63, 2.02)	1.28 (0.74, 2.21)	1.28 (0.73, 2.24)	1.34 (0.80, 2.24)	1.27 (0.60, 2.69)	1.39 (0.84, 2.31)	1.39 (0.84, 2.32)	1.41 (0.85, 2.33)	1.46 (0.86, 2.49)	1.56 (0.84, 2.88)	1.51 (0.91, 2.53)	1.75 (0.96, 3.21)
0.88 (0.49, 1.58)	AML	1.13 (0.79, 1.63)	1.13 (0.77, 1.66)	1.18 (0.87, 1.61)	1.12 (0.60, 2.11)	1.23 (0.92, 1.65)	1.23 (0.91, 1.67)	1.25 (0.93, 1.67)	1.29 (0.92, 1.82)	1.38 (0.87, 2.18)	1.34 (0.99, 1.82)	1.55 (1.00, 2.42)
0.78 (0.45, 1.35)	0.88 (0.61, 1.26)	RAD	1.00 (0.72, 1.38)	1.04 (0.82, 1.32)	0.99 (0.54, 1.80)	1.09 (0.87, 1.35)	1.09 (0.87, 1.36)	1.10 (0.89, 1.36)	1.14 (0.86, 1.50)	1.21 (0.80, 1.83)	1.18 (0.94, 1.49)	1.37 (0.92, 2.03)
0.78 (0.45, 1.37)	0.88 (0.60, 1.30)	1.00 (0.72, 1.39)	UST	1.05 (0.80, 1.37)	0.99 (0.54, 1.83)	1.09 (0.84, 1.40)	1.09 (0.84, 1.42)	1.10 (0.86, 1.42)	1.14 (0.84, 1.55)	1.22 (0.79, 1.88)	1.18 (0.91, 1.55)	1.37 (0.90, 2.08)
0.75 (0.45, 1.25)	0.85 (0.62, 1.15)	0.96 (0.76, 1.22)	0.96 (0.73, 1.26)	LEB	0.95 (0.54, 1.68)	1.04 (0.93, 1.17)	1.04 (0.92, 1.18)	1.05 (0.95, 1.17)	1.09 (0.89, 1.34)	1.16 (0.80, 1.68)	1.13 (0.99, 1.30)	1.31 (0.92, 1.86)
0.79 (0.37, 1.67)	0.89 (0.47, 1.67)	1.01 (0.56, 1.84)	1.01 (0.55, 1.86)	1.05 (0.60, 1.86)	FEZ	1.10 (0.63, 1.92)	1.10 (0.63, 1.93)	1.11 (0.64, 1.94)	1.15 (0.64, 2.07)	1.23 (0.63, 2.37)	1.19 (0.68, 2.10)	1.38 (0.72, 2.65)
0.72 (0.43, 1.19)	0.81 (0.60, 1.09)	0.92 (0.74, 1.14)	0.92 (0.71, 1.18)	0.96 (0.86, 1.08)	0.91 (0.52, 1.60)	DUP	1.00 (0.92, 1.09)	1.01 (0.97, 1.06)	1.05 (0.87, 1.26)	1.12 (0.78, 1.60)	1.09 (0.98, 1.20)	1.26 (0.90, 1.77)
0.72 (0.43, 1.19)	0.81 (0.60, 1.09)	0.92 (0.73, 1.15)	0.92 (0.71, 1.19)	0.96 (0.85, 1.09)	0.91 (0.52, 1.60)	1.00 (0.92, 1.09)	TRA	1.01 (0.94, 1.09)	1.05 (0.86, 1.27)	1.12 (0.78, 1.60)	1.09 (0.97, 1.22)	1.26 (0.89, 1.77)
0.71 (0.43, 1.17)	0.80 (0.60, 1.07)	0.91 (0.74, 1.12)	0.91 (0.71, 1.17)	0.95 (0.85, 1.05)	0.90 (0.51, 1.57)	0.99 (0.95, 1.03)	0.99 (0.92, 1.06)	Placebo	1.03 (0.87, 1.23)	1.10 (0.77, 1.57)	1.07 (0.98, 1.18)	1.24 (0.89, 1.74)
0.69 (0.40, 1.17)	0.77 (0.55, 1.09)	0.88 (0.67, 1.16)	0.88 (0.64, 1.19)	0.92 (0.75, 1.13)	0.87 (0.48, 1.56)	0.95 (0.80, 1.14)	0.96 (0.79, 1.16)	0.97 (0.81, 1.15)	CM310	1.07 (0.72, 1.58)	1.04 (0.85, 1.27)	1.20 (0.82, 1.76)
0.64 (0.35, 1.19)	0.73 (0.46, 1.15)	0.82 (0.55, 1.25)	0.82 (0.53, 1.27)	0.86 (0.59, 1.24)	0.82 (0.42, 1.58)	0.89 (0.63, 1.28)	0.90 (0.62, 1.29)	0.91 (0.64, 1.29)	0.94 (0.63, 1.39)	SPE	0.97 (0.68, 1.40)	1.13 (0.69, 1.84)
0.66 (0.40, 1.10)	0.75 (0.55, 1.01)	0.85 (0.67, 1.07)	0.84 (0.65, 1.10)	0.88 (0.77, 1.02)	0.84 (0.48, 1.48)	0.92 (0.83, 1.02)	0.92 (0.82, 1.04)	0.93 (0.85, 1.02)	0.96 (0.79, 1.18)	1.03 (0.71, 1.48)	NEM	1.16 (0.82, 1.64)
0.57 (0.31, 1.04)	0.64 (0.41, 1.00)	0.73 (0.49, 1.09)	0.73 (0.48, 1.11)	0.76 (0.54, 1.08)	0.72 (0.38, 1.39)	0.79 (0.57, 1.11)	0.79 (0.56, 1.12)	0.80 (0.58, 1.12)	0.83 (0.57, 1.21)	0.89 (0.54, 1.44)	0.86 (0.61, 1.22)	EBL

FIGURE 7 | League plots of adverse events (a), serious adverse events (b) and AE leading to discontinuation (c). Bold texts indicated that the comparison between the two groups was statistically significant. AML: Amltelimab, AST: Astegolimab, DUP: Dupilumab, EBL: Eblasakimab, FEZ: Fezakinumab, LEB: Lebrikizumab, NEM: Nemolizumab, RAD: Rademikibart, SPE: Spesolimab, TRA: Tralokinumab, UST: Ustekinumab.

From the perspective of SUCRA ranking, Spesolimab may be the best option to improve EASI-50. Rademikibart may be considered more effective than other interventions in improving EASI-75, while Dupilumab may be most beneficial for EASI-90. Dupilumab had better efficacy in improving NRS, EASI

and SCORAD scores. In terms of safety, Dupilumab may be safer than other monoclonal antibodies. However, the variability in data may lead to statistical uncertainty in SUCRA ranking; further analysis is needed when it is applied in clinical practice.

b

DUP	0.82 (0.12, 5.43)	1.16 (0.40, 3.37)	1.34 (0.33, 5.39)	1.66 (0.68, 4.04)	2.16 (1.08, 4.31)	2.32 (1.59, 3.38)	4.43 (0.37, 53.02)	5.94 (0.29, 120.9)	3.78 (1.46, 9.78)	6.36 (0.26, 154.0)	3.91 (1.62, 9.43)	6.08 (0.77, 47.94)
1.22 (0.18, 8.13)	CM310	1.41 (0.17, 11.66)	1.64 (0.17, 16.18)	2.03 (0.27, 15.37)	2.64 (0.38, 18.44)	2.84 (0.44, 18.15)	5.43 (0.25, 117.5)	7.26 (0.22, 245.3)	4.63 (0.60, 35.96)	7.78 (0.20, 305.0)	4.79 (0.64, 36.06)	7.43 (0.47, 116.4)
0.87 (0.30, 2.53)	0.71 (0.09, 5.83)	SPE	1.16 (0.22, 6.18)	1.44 (0.40, 5.20)	1.87 (0.59, 5.94)	2.01 (0.74, 5.47)	3.84 (0.27, 54.27)	5.14 (0.22, 120.3)	3.27 (0.87, 12.35)	5.50 (0.20, 152.1)	3.39 (0.94, 12.17)	5.26 (0.55, 50.62)
0.75 (0.19, 3.00)	0.61 (0.06, 6.01)	0.86 (0.16, 4.59)	RAD	1.24 (0.26, 5.91)	1.61 (0.37, 6.92)	1.73 (0.45, 6.60)	3.31 (0.20, 54.06)	4.43 (0.17, 117.2)	2.82 (0.57, 13.93)	4.74 (0.15, 147.3)	2.92 (0.61, 13.84)	4.53 (0.40, 51.59)
0.60 (0.25, 1.46)	0.49 (0.07, 3.71)	0.70 (0.19, 2.51)	0.81 (0.17, 3.84)	LEB	1.30 (0.48, 3.49)	1.39 (0.63, 3.11)	2.67 (0.20, 35.21)	3.57 (0.16, 78.99)	2.27 (0.70, 7.44)	3.83 (0.15, 100.1)	2.35 (0.76, 7.28)	3.65 (0.41, 32.45)
0.46 (0.23, 0.93)	0.38 (0.05, 2.65)	0.54 (0.17, 1.71)	0.62 (0.14, 2.68)	0.77 (0.29, 2.08)	TRA	1.08 (0.60, 1.92)	2.06 (0.17, 25.56)	2.75 (0.13, 57.93)	1.75 (0.62, 5.00)	2.95 (0.12, 73.62)	1.81 (0.68, 4.85)	2.82 (0.34, 23.29)
0.43 (0.30, 0.63)	0.35 (0.06, 2.26)	0.50 (0.18, 1.36)	0.58 (0.15, 2.21)	0.72 (0.32, 1.60)	0.93 (0.52, 1.66)	Placebo	1.91 (0.16, 22.21)	2.56 (0.13, 50.95)	1.63 (0.68, 3.90)	2.74 (0.12, 64.94)	1.69 (0.76, 3.74)	2.62 (0.34, 19.97)
0.23 (0.02, 2.70)	0.18 (0.01, 3.99)	0.26 (0.02, 3.69)	0.30 (0.02, 4.94)	0.37 (0.03, 4.95)	0.49 (0.04, 6.04)	0.52 (0.05, 6.07)	AML	1.34 (0.03, 64.03)	0.85 (0.06, 11.51)	1.43 (0.03, 78.57)	0.88 (0.07, 11.62)	1.37 (0.06, 33.08)
0.17 (0.01, 3.43)	0.14 (0.00, 4.65)	0.19 (0.01, 4.56)	0.23 (0.01, 5.98)	0.28 (0.01, 6.19)	0.36 (0.02, 7.64)	0.39 (0.02, 7.77)	0.75 (0.02, 35.71)	FEZ	0.64 (0.03, 14.35)	1.07 (0.01, 83.31)	0.66 (0.03, 14.54)	1.02 (0.03, 38.01)
0.26 (0.10, 0.68)	0.22 (0.03, 1.68)	0.31 (0.08, 1.15)	0.35 (0.07, 1.75)	0.44 (0.13, 1.44)	0.57 (0.20, 1.62)	0.61 (0.26, 1.47)	1.17 (0.09, 15.83)	1.57 (0.07, 35.39)	NEM	1.68 (0.06, 44.81)	1.03 (0.32, 3.37)	1.61 (0.18, 14.65)
0.16 (0.01, 3.81)	0.13 (0.00, 5.04)	0.18 (0.01, 5.02)	0.21 (0.01, 6.55)	0.26 (0.01, 6.84)	0.34 (0.01, 8.46)	0.36 (0.02, 8.63)	0.70 (0.01, 38.20)	0.93 (0.01, 72.63)	0.59 (0.02, 15.84)	AST	0.62 (0.02, 16.07)	0.96 (0.02, 41.03)
0.26 (0.11, 0.62)	0.21 (0.03, 1.57)	0.30 (0.08, 1.06)	0.34 (0.07, 1.63)	0.42 (0.14, 1.31)	0.55 (0.21, 1.47)	0.59 (0.27, 1.31)	1.13 (0.09, 14.92)	1.52 (0.07, 33.49)	0.97 (0.30, 3.14)	1.63 (0.06, 42.46)	UST	1.55 (0.18, 13.75)
0.16 (0.02, 1.30)	0.13 (0.01, 2.11)	0.19 (0.02, 1.83)	0.22 (0.02, 2.51)	0.27 (0.03, 2.43)	0.35 (0.04, 2.93)	0.38 (0.05, 2.91)	0.73 (0.03, 17.62)	0.98 (0.03, 36.30)	0.62 (0.07, 5.67)	1.05 (0.02, 44.96)	0.64 (0.07, 5.70)	EBL

FIGURE 7 | (Continued)

c

TRA	1.08 (0.19, 6.20)	1.47 (0.84, 2.57)	1.48 (0.73, 2.99)	1.76 (0.48, 6.44)	1.83 (0.73, 4.55)	2.15 (0.33, 13.84)	2.11 (0.81, 5.49)	2.85 (0.23, 35.46)	3.76 (0.18, 78.89)	4.03 (0.16, 100.2)	3.41 (0.41, 28.25)	4.59 (0.46, 45.80)
0.93 (0.16, 5.34)	RAD	1.36 (0.26, 7.15)	1.37 (0.25, 7.59)	1.63 (0.22, 12.42)	1.69 (0.28, 10.32)	1.99 (0.18, 22.61)	1.96 (0.31, 12.20)	2.64 (0.14, 51.24)	3.49 (0.11, 106.6)	3.74 (0.11, 133.1)	3.17 (0.23, 43.78)	4.25 (0.26, 68.56)
0.68 (0.39, 1.19)	0.73 (0.14, 3.85)	Placebo	1.01 (0.66, 1.54)	1.20 (0.37, 3.86)	1.24 (0.61, 2.55)	1.46 (0.25, 8.64)	1.44 (0.66, 3.12)	1.94 (0.17, 22.65)	2.56 (0.13, 50.95)	2.74 (0.12, 64.94)	2.32 (0.30, 17.82)	3.12 (0.33, 29.07)
0.68 (0.33, 1.37)	0.73 (0.13, 4.04)	0.99 (0.65, 1.52)	DUP	1.19 (0.34, 4.13)	1.24 (0.54, 2.85)	1.45 (0.23, 9.03)	1.43 (0.59, 3.46)	1.93 (0.16, 23.36)	2.55 (0.12, 52.20)	2.73 (0.11, 66.43)	2.31 (0.29, 18.51)	3.10 (0.32, 30.09)
0.57 (0.16, 2.07)	0.61 (0.08, 4.65)	0.83 (0.26, 2.68)	0.84 (0.24, 2.91)	SPE	1.04 (0.26, 4.09)	1.22 (0.15, 10.22)	1.20 (0.29, 4.87)	1.62 (0.11, 24.59)	2.14 (0.09, 52.97)	2.29 (0.08, 66.75)	1.94 (0.19, 20.29)	2.60 (0.21, 32.32)
0.55 (0.22, 1.36)	0.59 (0.10, 3.59)	0.80 (0.39, 1.65)	0.81 (0.35, 1.87)	0.96 (0.24, 3.80)	NEM	1.17 (0.17, 7.99)	1.16 (0.40, 3.33)	1.56 (0.12, 20.20)	2.06 (0.10, 44.64)	2.21 (0.09, 56.63)	1.87 (0.22, 16.21)	2.51 (0.24, 26.19)
0.47 (0.07, 3.00)	0.50 (0.04, 5.71)	0.68 (0.12, 4.05)	0.69 (0.11, 4.28)	0.82 (0.10, 6.89)	0.85 (0.13, 5.79)	AML	0.98 (0.14, 6.84)	1.33 (0.06, 27.57)	1.75 (0.05, 56.86)	1.88 (0.05, 70.81)	1.59 (0.11, 23.76)	2.14 (0.12, 37.06)
0.47 (0.18, 1.23)	0.51 (0.08, 3.19)	0.70 (0.32, 1.51)	0.70 (0.29, 1.70)	0.83 (0.21, 3.39)	0.87 (0.30, 2.49)	1.02 (0.15, 7.07)	LEB	1.35 (0.10, 17.77)	1.78 (0.08, 39.17)	1.91 (0.07, 49.67)	1.62 (0.18, 14.31)	2.17 (0.20, 23.08)
0.35 (0.03, 4.38)	0.38 (0.02, 7.35)	0.52 (0.04, 6.04)	0.52 (0.04, 6.30)	0.62 (0.04, 9.42)	0.64 (0.05, 8.32)	0.75 (0.04, 15.68)	0.74 (0.06, 9.77)	CM310	1.32 (0.03, 63.53)	1.42 (0.03, 77.95)	1.20 (0.05, 29.25)	1.61 (0.06, 44.62)
0.27 (0.01, 5.57)	0.29 (0.01, 8.75)	0.39 (0.02, 7.77)	0.39 (0.02, 8.05)	0.47 (0.02, 11.60)	0.49 (0.02, 10.51)	0.57 (0.02, 18.48)	0.56 (0.03, 12.31)	0.76 (0.02, 36.32)	FEZ	1.07 (0.01, 83.31)	0.91 (0.02, 33.82)	1.22 (0.03, 50.86)
0.25 (0.01, 6.17)	0.27 (0.01, 9.52)	0.36 (0.02, 8.63)	0.37 (0.02, 8.93)	0.44 (0.01, 12.75)	0.45 (0.02, 11.63)	0.53 (0.01, 20.06)	0.52 (0.02, 13.61)	0.71 (0.01, 38.85)	0.93 (0.01, 72.63)	AST	0.85 (0.02, 36.50)	1.14 (0.02, 54.67)
0.29 (0.04, 2.42)	0.32 (0.02, 4.37)	0.43 (0.06, 3.30)	0.43 (0.05, 3.47)	0.52 (0.05, 5.40)	0.54 (0.06, 4.64)	0.63 (0.04, 9.39)	0.62 (0.07, 5.47)	0.83 (0.03, 20.32)	1.10 (0.03, 41.11)	1.18 (0.03, 50.90)	EBL	1.34 (0.07, 27.58)
0.22 (0.02, 2.18)	0.24 (0.01, 3.79)	0.32 (0.03, 2.99)	0.32 (0.03, 3.13)	0.38 (0.03, 4.77)	0.40 (0.04, 4.16)	0.47 (0.03, 8.12)	0.46 (0.04, 4.89)	0.62 (0.02, 17.19)	0.82 (0.02, 34.27)	0.88 (0.02, 42.25)	0.74 (0.04, 15.29)	UST

FIGURE 7 | (Continued)

Acknowledgments

We thank all authors for their important contributions.

Funding

The authors have nothing to report.

Disclosure

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting Information.